Clinical and Physiological Evaluation of Surgicover: A Specialized Amino Acid and Micronutrient Formulation in Perioperative and Postoperative Recovery

Executive Summary

The metabolic and immunological stress response to major surgical trauma presents a significant clinical challenge in perioperative medicine. Surgical interventions trigger a predictable neuroendocrine cascade characterized by the hypersecretion of counter-regulatory hormones—such as cortisol, catecholamines, and glucagon—alongside a massive systemic release of pro-inflammatory cytokines. This hypermetabolic state leads to accelerated glycogenolysis, lipolysis, and skeletal muscle proteolysis, resulting in rapid nitrogen loss and negative nitrogen balance. Malnutrition and postoperative underfeeding act as independent risk factors that compound these metabolic derangements, directly translating to elevated rates of wound dehiscence, anastomotic leaks, surgical site infections, and prolonged hospital stays.
To counteract this catabolic state, modern clinical protocols, such as Enhanced Recovery After Surgery (ERAS), emphasize early enteral feeding and targeted immunonutrition. This clinical whitepaper examines the physiological mechanisms of Surgicover, a specialized formulation containing L-Arginine, L-Leucine, Soy Protein Isolate, and essential vitamins and minerals. It details how these components interact at a cellular level to preserve lean muscle mass, optimize microvascular perfusion, modulate inflammatory pathways, and accelerate tissue regeneration across twelve diverse surgical departments.

Nutritional Profile and Specifications of the Specialized Formulation

The therapeutic efficacy of Surgicover is built upon a balanced distribution of macronutrients and targeted micronutrients designed to satisfy the elevated metabolic demands of the surgical patient. The exact nutritional values per serving and per 100g are detailed in the table below.

Nutrient	Unit	Concentration per 100g	Concentration per 20g Serving	% RDA (per Serving)
Energy	Kcal	378.20	75.64	----------
Total Carbohydrate	gm	79.70	15.94	----------
Added Sugar (Sucrose)	gm	0.00	0.00	----------
Protein (30%)	gm	30.00	6.00	----------

Nutrient	Unit	Concentration per 100g	Concentration per 20g Serving	% RDA (per Serving)
Fat	gm	0.96	0.19	----------
Saturated Fats	mg	0.55	0.11	----------
L-Arginine	mg	1000.00	200.00	----------
L-Leucine	mg	500.00	100.00	----------
Vitamin B1	mg	3.33	0.60	37.00%
Vitamin B2	mg	3.66	0.73	37.00%
Niacinamide	mg	40.00	8.00	44.00%
Vitamin B12	mcg	1.60	0.32	27.00%
Pantothenic Acid	mg	3.33	0.67	13.00%
Vitamin D	IU	300.00	60.00	10.00%
Vitamin B6	mg	1.60	0.32	13.00%
Folic Acid	mcg	760.00	152.00	27.00%
Vitamin C	mg	160.00	32.00	40.00%
Manganese	mg	2.50	0.50	12.00%
Iron	mg	12.00	2.40	7.00%
Copper	mg	1.60	0.32	19.00%
Calcium	gm	1.616	0.323 (323 mg)	32.33%
Phosphorous	gm	1.266	0.253 (253 mg)	25.33%

Ingredients: Soya Protein Isolate, Defatted Soya Flour, Skimmed Milk Powder, Sucralose, L-Arginine, L-Leucine, Di-basic Calcium Phosphate, Cyanocobalamin, Thiamine Mononitrate, Riboflavin, Niacinamide, Pantothenic acid, Pyridoxine Hydrochloride, Calcium D-pantothenate, Vitamin D, Folic acid, Ascorbic acid, Ferric Ammonium Citrate, Manganese Sulphate, Copper Sulphate, Preservative (INS211). Contains zero added sugar (sucrose) and is sweetened with sucralose, presenting a heart-healthy and glycemic-stable profile under a dry fruits flavor.

Molecular Mechanisms of the Active Constituents

L-Arginine and the Nitric Oxide-Arginase Pathways

L-Arginine is a conditionally indispensable amino acid containing 32\% nitrogen, making it a critical nitrogen carrier during trauma and surgical stress. It serves as the primary substrate for two competing intracellular enzyme families: nitric oxide synthases (\text{NOS}) and arginases. The \text{NOS} pathway catalyzes the oxidation of L-Arginine to L-Citrulline and nitric oxide (\text{NO}) according to the following biochemical reaction :
\text{L-Arginine} + \text{O}_2 + \text{NADPH} + \text{H}^+ \xrightarrow{\text{NOS}} \text{L-Citrulline} + \text{NO} + \text{NADP}^+
Nitric oxide functions as a potent pleiotropic regulator of vascular tone and tissue microperfusion. By diffusing into vascular smooth muscle, \text{NO} binds to soluble guanylate cyclase (\text{sGC}), triggering the conversion of guanosine triphosphate (\text{GTP}) to cyclic guanosine monophosphate (\text{cGMP}). This activates protein kinase G (\text{PKG}), resulting in cellular calcium extrusion, smooth muscle relaxation, and vasodilation. Under stress, endothelial cell dysfunction often compromises endogenous \text{NO} release, impairing tissue perfusion and driving local hypoxia. Supplemental L-Arginine maintains microvascular flow, optimizes oxygen delivery, and stimulates growth hormone and insulin-like growth factor-1

(\text{IGF}-1), accelerating systemic tissue regeneration.
Simultaneously, the arginase pathway metabolizes L-Arginine to L-Ornithine and urea.
L-Ornithine is converted via ornithine aminotransferase into proline, an essential amino acid that undergoes post-translational hydroxylation to form hydroxyproline. Hydroxyproline is the primary structural component required for collagen synthesis and wound healing.

L-Leucine, Complete Protein Sparing, and mTORC1 Activation

The preservation of skeletal muscle during postoperative immobilization relies heavily on the mechanistic target of rapamycin complex 1 (\text{mTORC}1) signaling cascade. L-Leucine serves as a direct intracellular nutrient sensor and activator of \text{mTORC}1. Upon activation, \text{mTORC}1 phosphorylates its downstream effectors, p70 ribosomal S6 kinase 1 (\text{p70S6K}1) and eukaryotic translation initiation factor 4E-binding protein 1 (\text{4E-BP}1), initiating muscle protein synthesis (\text{MPS}).
However, studies show that supplementing with L-Leucine alone is insufficient to prevent disuse-induced muscle loss. While L-Leucine acts as the molecular "on-switch" for translation, the process of protein synthesis cannot be sustained without a complete pool of all essential amino acids (\text{EAAs}). If the remaining \text{EAAs} are not provided exogenously, the intracellular signaling pathway stalls, and the body breaks down endogenous muscle tissue to acquire the missing substrates, rendering the anabolic stimulus ineffective.
Surgicover solves this clinical bottleneck by combining free L-Leucine with high-purity Soy Protein Isolate (\text{SPI}), Defatted Soya Flour, and Skimmed Milk Powder. This matrix provides both the molecular trigger (\text{mTORC}1 activation) and the complete essential amino acid substrate spectrum required to sustain protein translation, reverse negative nitrogen balance, and mitigate muscle catabolism.

Soy Protein Isolate as a Complete Plant Protein and Gut Barrier Preserver

Soy Protein Isolate (\text{SPI}) is a complete, high-quality plant-based protein with a Protein Digestibility-Corrected Amino Acid Score (\text{PDCAAS}) of 0.95 to 1.02, making its quality comparable to milk and egg proteins. SPI naturally contains high concentrations of branched-chain amino acids (\text{BCAAs}), glutamine, and L-Arginine, which are critical for metabolic regulation and tissue repair.
In addition to supporting systemic nitrogen balance, dietary SPI preserves gut barrier function during surgical stress. Major abdominal surgeries or trauma can compromise the intestinal mucosa, leading to tight junction failure and subclinical protein-losing enteropathy. Research shows that SPI downregulates mucosal inflammatory cytokines—such as interleukin-4 (\text{IL}-4) and interleukin-13 (\text{IL}-13)—and stabilizes the expression of polymeric immunoglobulin receptors and mucosal mucins. This helps prevent the translocation of luminal pathogens, maintains gut barrier integrity, and reduces systemic inflammatory responses.

Department-Specific Clinical Papers and Whitepapers

1. Orthopedic Surgery: Mitigation of Disuse Atrophy and Connective Tissue Repair

Postoperative immobilization and non-weight-bearing restrictions after major orthopedic reconstructions—including anterior cruciate ligament reconstruction (\text{ACLR}), total hip arthroplasty (\text{THA}), and total knee arthroplasty (\text{TKA})—induce rapid skeletal muscle atrophy, particularly within the quadriceps. This disuse-induced atrophy is driven by a profound reduction in basal and postprandial muscle protein synthesis (\text{MPS}) rates, often compounded by age-related anabolic resistance.
To overcome this state of anabolic resistance and promote physical recovery, the administration of high-quality protein is mandatory. While isolated L-Leucine is cleared rapidly and cannot maintain translation independently, the combination of free L-Leucine with complete Soy Protein Isolate provides both the molecular trigger (\text{mTORC}1 activation) and the necessary essential amino acid substrate pool to sustain \text{MPS} and prevent lean mass loss. Clinical trials demonstrate that protein supplementation mitigates muscle atrophy post-immobilization, correlating with improved functional measures and quicker achievement of rehabilitation benchmarks.
Concurrently, the high demand for structural tissue repair in ligaments, tendons, and joint capsules requires accelerated collagen synthesis. L-Arginine, through its conversion to proline, works alongside Vitamin C and Zinc to drive fibroblast proliferation and collagen matrix deposition, improving graft incorporation and joint stability. Additionally, bone tissue mineralization during osteoblast-mediated skeletal repair is supported by cholecalciferol (Vitamin D), which regulates calcium and phosphate transport across the osteoid matrix.

2. Urology and Pelvic Reconstruction: Endothelial Recovery and Cavernous Nitric Oxide Pathway Rehabilitation

Following radical prostatectomy, patients frequently experience transient or prolonged erectile dysfunction due to intraoperative traction, thermal injury, or dissection of the cavernous nerves. Even when nerve-sparing techniques are employed, postoperative neuropraxia leads to a lack of endogenous nitric oxide release within the penile vasculature, causing chronic hypoxia of the cavernous tissue. Over time, this hypoxia induces smooth muscle apoptosis, fibrotic replacement of the cavernous tissue, and structural veno-occlusive dysfunction.
The specialized formulation addresses this pathobiology by targeting the endothelial nitric oxide pathway during the critical rehabilitation window. High-dose L-Arginine serves as the direct substrate for endothelial nitric oxide synthase (\text{eNOS}), promoting steady, non-adrenergic vasodilation within the internal pudendal arteries and cavernous tissue. This continuous microvascular perfusion preserves oxygenation, limits collagenous degeneration, and maintains cavernous smooth muscle elasticity while awaiting nerve regeneration.
This vascular recovery acts synergistically with daily low-dose phosphodiesterase-5 inhibitors (\text{PDE}5\text{Is}) and vacuum erection devices (\text{VED}). Additionally, the mucosal anastomosis of the bladder neck and urethra relies on optimal local microperfusion and zinc-dependent protein synthesis to ensure rapid, leak-free healing, reducing the duration of postoperative catheterization.

3. General Surgery: Nitrogen Sparing, Fascial Closure, and Immunonutrition in Abdominal Trauma

Major abdominal surgeries, including laparotomies, bowel resections, and complex hernia repairs, trigger a severe neuroendocrine stress response that causes rapid protein catabolism

and negative nitrogen balance. This catabolic state is a primary driver of wound dehiscence and surgical site infections (SSIs), as the body prioritizes visceral organ function over abdominal wall fascial repair.
To support the high metabolic demands of general surgery patients, clinical guidelines recommend a protein intake of 1.2 to $2.0\text{ g/kg ideal body weight/day}$. Enteral administration of the specialized SPI-leucine formulation meets these elevated protein requirements, limiting skeletal muscle proteolysis and providing the necessary nitrogen balance to fuel fascial healing.
Concurrently, L-Arginine acts as an immunomodulator, enhancing T-lymphocyte proliferation and function, which reduces infectious complications and shortens the length of hospital stays. Vitamin C and Zinc serve as essential cofactors for the synthesis and cross-linking of the fascial collagen matrix, ensuring robust wound tensile strength and minimizing the risk of postoperative hernia formation or abdominal wound dehiscence.

4. Gynecology and Obstetrics: Post-Hysterectomy Pelvic Floor Restoration and Soft-Stool Constipation Management

Major gynecological procedures, such as total hysterectomies and pelvic organ prolapse reconstructions, require extensive dissection of the pelvic fascia and ligaments, as well as incisions through the vaginal vault. The pelvic floor musculature, including the levator ani and coccygeus muscles, must heal while subjected to continuous intra-abdominal pressure. Furthermore, postoperative constipation and subsequent straining pose a mechanical threat to the newly sutured pelvic floor and vaginal cuff, risking vault dehiscence or early prolapse recurrence.
To support pelvic floor recovery, the formulation utilizes the complete amino acid profile of SPI and L-Leucine to promote myofibrillar repair of the pelvic diaphragm. L-Arginine, through its conversion to proline, and Vitamin C accelerate collagen remodeling within the endopelvic fascia and uterosacral ligaments, enhancing their structural tensile strength.
To protect the healing surgical site from mechanical strain, SPI provides a highly digestible, plant-based protein source. Unlike dense animal proteins, SPI does not delay gastric emptying or prolong transit time, reducing the incidence of postoperative constipation. This dietary support, combined with targeted, low-impact pelvic floor exercises (Kegel exercises), accelerates functional recovery and maintains pelvic organ support.

5. Surgical Oncology: Counteracting Cancer Cachexia, Muscle Proteolysis, and Systemic Inflammation

Oncological patients undergoing major surgical resections often present with pre-existing cancer cachexia, a complex metabolic syndrome characterized by systemic inflammation, involuntary skeletal muscle wasting, and profound negative nitrogen balance. Tumor-secreted factors and host pro-inflammatory cytokines, such as interleukin-1 beta ($\text{IL}-1\beta$) and tumor necrosis factor-alpha ($\text{TNF}-\alpha$), drive hypermetabolism and accelerate the degradation of muscle proteins. This state of malnutrition impairs the patient's immune system and reduces tolerance to surgery, chemotherapy, and radiation.
In this highly catabolic state, the combination of free L-Leucine and SPI directly targets the deregulated protein metabolism of cachexia. Leucine acts as an anabolic signal to stimulate the $\text{mTORC}1$ pathway, counteracting tumor-induced muscle proteolysis. SPI provides a

balanced, non-tumor-promoting plant-based protein source that restores plasma amino acid levels and supplies the nitrogen required to support hepatic acute-phase protein synthesis. Additionally, the immunomodulating properties of L-Arginine enhance T-cell receptor expression, counteracting the immunosuppressive effects of surgical trauma and oncological therapies. This nutritional prehabilitation preserves lean mass, improves immunological competence, and directly reduces postoperative mortality and infectious complications.

6. Gastroenterology and Gastrointestinal Surgery: Prevention of Splanchnic Capillary Leak and Postoperative Hypoalbuminemia

Postoperative hypoalbuminemia is a frequent and serious complication following major gastrointestinal procedures, such as pancreaticoduodenency or colectomy, affecting up to 20\% to 40\% of patients. This acute drop in serum albumin is primarily driven by a form of transient, subclinical protein-losing enteropathy. Surgical trauma and the release of inflammatory cytokines compromise the structural integrity of the small intestine's mucosal barrier. The intestinal lining, which covers a massive surface area of 250\text{ m}^2, develops defective tight junctions between epithelial cells. Consequently, the gut switches from an absorptive surface to an excretory leak, allowing serum proteins, such as albumin and immunoglobulins, to extravasate down their concentration gradients into the intestinal lumen.

The specialized SPI formulation helps prevent this pathology by preserving the intestinal mucosal barrier. SPI has been shown to downregulate pro-inflammatory pathways and modulate the expression of polymeric immunoglobulin receptors and intestinal mucins to optimal physiological levels, preventing hyperpermeability.

By maintaining the integrity of these intercellular junctions, the formulation prevents the retrograde loss of serum albumin, limiting interstitial edema, preserving vascular volume, and reducing the risk of third-spacing, prolonged ileus, and infectious complications.

7. Neurosurgery: Metabolic Support for Craniotomy and Neurogenic Recovery

Neurosurgical procedures, such as craniotomies for tumor resections or interventions for traumatic brain injury (\text{TBI}), trigger a massive neuroendocrine stress response. The resulting hypermetabolic state can elevate resting energy expenditure by 87\% to 200\% above baseline, leading to rapid muscle proteolysis. However, while systemic metabolism is highly catabolic, cerebral glucose metabolism may be globally or regionally depressed, risking secondary ischemic neuronal injury.

The formulation addresses these neurological metabolic demands through multiple pathways. High-purity SPI and L-Leucine supply the complete amino acid profile required to offset neurogenic muscle wasting without provoking the rapid blood glucose fluctuations that can exacerbate secondary brain injury.

L-Arginine acts as the direct precursor for nitric oxide, which optimizes cerebral microperfusion, helps maintain the blood-brain barrier, and protects tissues from localized ischemia. Concurrently, Zinc acts as an essential modulator of synaptic plasticity and neurotransmission, working alongside B-complex vitamins to promote axonal repair and cognitive recovery.

8. Vascular Surgery: Mitigation of Ischemia/Reperfusion Injury and

Graft Patency

In vascular surgery, reestablishing blood flow to ischemic limbs or organs—such as during peripheral arterial bypass grafting or carotid endarterectomy—triggers ischemia/reperfusion (I/R) injury within the skeletal muscle and endothelium. Under ischemic conditions, local L-Arginine pools are depleted due to the failure of the ATP-dependent argininosuccinate synthetase recycling pathway. When constitutive nitric oxide synthase (cNOS) is activated in an L-Arginine-starved environment during reperfusion, it undergoes conformational uncoupling. Instead of producing protective NO, the uncoupled enzyme transfers electrons directly to oxygen, generating highly toxic superoxide radicals ($\text{O}_2^{\bullet-}$) according to the following biochemical pathway :

$$\text{O}_2 + \text{NADPH} \xrightarrow{\text{uncoupled cNOS}} \text{O}_2^{\bullet-} + \text{NADP}^+ + \text{H}^+$$

These superoxide radicals react with residual NO to form peroxynitrite (OONO^-), a potent vasoconstrictor and cytotoxin that causes endothelial necrosis and microvascular edema :

$$\text{NO} + \text{O}_2^{\bullet-} \xrightarrow{} \text{OONO}^-$$

Supplemental L-Arginine prevents cNOS uncoupling by replenishing the intracellular substrate pool. This shifts the cellular kinetics back toward the synthesis of vascular nitric oxide, neutralizing superoxide generation and reducing reperfusion-induced vasoconstriction and microvascular edema. By maintaining endothelial cell integrity and reducing vascular thrombosis, the formulation preserves the microcirculation, improves graft and arteriovenous fistula patency, and enhances overall limb salvage rates.

9. Cardiothoracic Vascular Surgery (CTVS): Sternal Osteogenesis, Fluid Volume Restriction, and Heart-Healthy Profiles

A median sternotomy involves a major incision through the sternum, creating a complex bone fracture in a hypermobile chest wall. Sternal bone healing is a challenging process; chest wall instability directly impairs the healing of adjacent soft tissues and increases the risk of deep sternal wound infections or mediastinitis. Furthermore, preoperative Vitamin D (calcidiol) deficiency is highly prevalent in cardiac surgery patients (affecting up to 70% of the surgical population) and is strongly associated with bone loss, impaired osteoblast function, and increased postoperative cardiovascular and infectious complications.

Standardized supplementation with cholecalciferol (Vitamin D) to achieve optimal calcidiol levels ($\ge 80\text{ nmol/L}$) is critical to accelerate osteogenesis. Calcidiol binds to the nuclear Vitamin D receptor on osteoblasts, promoting renal calcium retention, intestinal calcium absorption, and direct mineralization of the sternal bone matrix. Clinical trials show that elevating calcidiol levels significantly improves sternal bone healing and reduces sternal instability complications.

Co-administered Calcium provides the raw mineral substrate, while SPI-derived amino acids, Vitamin C, and Zinc catalyze the synthesis of the collagenous osteoid matrix, accelerating chest wall stability. Additionally, because cardiac patients typically accumulate 10 to 20 lbs of fluid post-surgery, strict sodium and fluid restrictions are necessary. Surgicover contains zero added sugar (sucrose) and extremely low fat (0.19g per serve), making it highly compatible with a heart-healthy diet.

10. Plastic and Reconstructive Surgery: Free Flap Microperfusion and

Wound Remodeling Mechanics

In plastic and reconstructive surgery, the success of microvascular free tissue transfers and skin grafts relies on maintaining adequate microvascular perfusion to prevent marginal necrosis, venous congestion, and flap loss. Following the surgical incision, the wound goes through a delicate remodeling phase where collagen synthesis must be balanced: insufficient collagen deposition leads to weak scars, while prolonged inflammation can result in hypertrophic scarring or keloids.

The formulation supports flap survival and scar optimization through multiple pathways.

L-Arginine optimizes free flap survival by promoting local \text{NO}-mediated vasodilation, increasing microvascular capillary perfusion and nutrient delivery. Simultaneously, the arginase pathway metabolizes arginine to ornithine, proline, and polyamines, which are the primary structural building blocks for collagen synthesis and tissue remodeling.

Vitamin C and Iron act as obligate cofactors for prolyl-4-hydroxylase, promoting correct cross-linking of the dermal extracellular matrix. Vitamin A is included to regulate the early inflammatory phase, promoting controlled angiogenesis and keratinocyte migration to ensure aesthetic, flat scar formation and reduce the incidence of postoperative fistulas.

11. Otorhinolaryngology (ENT) Surgery: Non-Abrasive Pharyngeal Recovery and Bleeding Prevention post-Tonsillectomy

Major pharyngeal surgeries and tonsillectomies result in severe postoperative pain (odynophagia), which restricts solid food intake for up to two weeks. This predisposing factor often leads to dehydration and acute weight loss. During recovery, the surgical sites are covered by a delicate, protective fibrinous scab. Consuming abrasive, hot, or sharp foods can mechanically scratch these scabs, triggering delayed postoperative hemorrhage, which typically peaks between postoperative days 4 and 10.

This formulation provides complete, high-density nutrition in a smooth, low-viscosity liquid vehicle that bypasses the need for mastication. SPI is highly soluble, bland, and non-acidic, making it non-irritating to the raw pharyngeal mucosa.

Providing the complete amino acid spectrum of SPI paired with free L-Leucine preserves systemic nitrogen balance and limits lean mass loss during the liquid-only recovery window. Zinc and Vitamin C accelerate mucosal re-epithelialization, hastening scab maturation and sloughing, which directly lowers the risk of delayed bleeding.

12. Dental and Maxillofacial Surgery: Sparing Masseter Muscle Atrophy during Jaw Fixation

Patients recovering from orthognathic surgery, complex jaw fractures, or jaw tumor resections often undergo maxillomandibular fixation (jaw wiring) or are placed on strict non-chewing restrictions for 4 to 8 weeks. Consuming adequate calories and protein through a liquid-only diet is a major challenge, frequently leading to profound muscle wasting, particularly of the masticatory muscles like the masseter. Mechanical trauma from solid particles can also cause hardware failure or surgical site infections.

The specialized liquid formulation is easily swallowed through the gaps in wired teeth or orthodontic elastics. The combination of SPI and L-Leucine directly counteracts the disuse atrophy of the masseter and temporalis muscles by keeping the \text{mTORC}1 pathway continuously active.

Calcium, phosphorous, and Vitamin D drive bone healing and callus formation at osteotomy or fracture sites. Vitamin A and Vitamin C support the integrity of the fragile oral mucosa and gums around dental implants and surgical hardware, accelerating epithelial closure and reducing the risk of implant exposure or osteomyelitis.

Comparative Analysis of Surgicover against Standard Polymeric ONS

The table below contrasts the distinct molecular targets, formulation properties, and clinical utility of Surgicover against generic polymeric oral nutritional supplements (ONS).

Parameters	Surgicover Specialized Formulation	Generic Polymeric ONS	Clinical Significance
Protein Source	Soya Protein Isolate + Defatted Soya Flour + Skimmed Milk Powder	Casein, Whey, or whole milk solids	SPI has high digestibility (PDCAAS ~ 1.0) and downregulates intestinal inflammatory cytokines, reducing the risk of protein-losing enteropathy.
Added Amino Acids	High-purity L-Arginine (1000 mg/100g) + L-Leucine (500 mg/100g)	Traces of standard profile amino acids only	High-dose L-Arginine optimizes vascular \text{NO} synthesis, while L-Leucine triggers \text{mTORC}1 to preserve skeletal muscle mass.
Glycemic Control	Zero added sugar (Sucrose); sweetened with Sucralose	High sucrose or corn starch content	Prevents hyperglycemia, which impairs neutrophil function and worsens surgical wound healing.
Fat Content	Very low fat (0.19\text{g} per serve) and low saturated fats (0.11\text{mg})	Higher lipid density (2.0\text{g} - 5.0\text{g} per serve)	Supports heart-healthy diets and matches strict recovery protocols for CTVS and vascular patients.
Mucosal Tolerability	Completely smooth, non-acidic, dry fruits flavored	Acidic or high-osmolality fruit flavors	Avoids pharyngeal mucosal irritation, improving compliance in ENT and oral surgery patients.

Clinical Protocols for Administration and Timing

Preoperative Prehabilitation Phase

Clinicians should initiate Surgicover supplementation at least seven to ten days prior to major elective surgeries, particularly in patients identified as having high nutritional risk or pre-existing sarcopenia. Preoperative administration of specific formulas enriched with L-Arginine and complete protein has been shown to optimize tissue nitrogen levels, stabilize immune responses, and reduce the rate of postoperative infectious complications. The recommended dosage during this phase is one 20\text{g} scoop dissolved in 150\text{ml} of lukewarm water or milk, administered twice daily, which delivers 400\text{mg} of L-Arginine and 200\text{mg} of L-Leucine per day.

Postoperative Sparing Phase

Following the return of bowel function, early enteral feeding should be established immediately. To counteract hypermetabolic muscle catabolism and maintain nitrogen balance during the high-risk muscle loss period, the patient should consume one serving of Surgicover two to three times daily for a minimum of two to four weeks postoperatively.
In patients undergoing physical therapy or active rehabilitation, administering Surgicover within twenty to thirty minutes following exercise optimizes \text{mTORC}1-mediated muscle protein synthesis. For patients with restricted chewing capabilities (such as those in ENT, dental, and maxillofacial surgery), the liquid vehicle can be easily administered via a cup, syringe, or feeding tube, ensuring continuous nutritional support without causing mechanical strain to raw healing tissues.

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